Polar Graphs (8.2) Name ______________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 5 # ________ The graph of $r = 9$ is a circle. State its radius.	ANSWER: 10 # ________ Identify the type of curve: $r = 3 + 3\cos\theta$.
ANSWER: 0 # ________ Evaluate r at $\theta = \frac{\pi}{2}$ for $r = 5\sin\theta$.	ANSWER: 2 # ________ Evaluate r at $\theta = \frac{\pi}{2}$ for $r = 2\cos\theta$.

ANSWER: 9 # ________ The graph of $r = 10\cos\theta$ is a circle. State its diameter.	ANSWER: cardioid # ________ Identify the type of curve: $r = 2 + 5\sin\theta$.
ANSWER: 8 # ________ Evaluate r at $\theta = 0$ for $r = 1 + \cos\theta$.	ANSWER: 4 # ________ State the maximum value of r for $r = 8\sin\theta$.
ANSWER: 3 # ________ How many petals does the rose $r = 4\sin(2\theta)$ have?	ANSWER: limaçon # ________ How many petals does the rose $r = 6\cos(3\theta)$ have?